

Akash Gogate

Princeton, NJ • 609-751-2597 • akash.gogate@gmail.com • agogate@wisc.edu • [linkedin.com/in/akashgogate](https://www.linkedin.com/in/akashgogate) • github.com/AkashGogate

EDUCATION

University of Wisconsin at Madison

Graduating: May 2028

Bachelor of Science in Computer Science • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Coursework: Cryptography, Advanced Algorithms, Machine Organization and Programming, Intro to Computer Engineering, Data Structures, Artificial Intelligence, Deep Learning, Computer Science Capstone, AP Computer Science A, AP Statistics, Game Development, Web Development

TECHNOLOGIES

Languages: Java, Python, TypeScript, Node.js, R, Rust, JavaScript, C#, C++, SQL, Linux/Bash

Systems & Data: Kafka, Kubernetes, Docker, MongoDB, PostgreSQL, SQLite, Redis, Supabase, REST APIs, FastAPI, GraphQL, NoSQL, Microservices, Low-Latency Systems, High-Performance Systems, Financial Data Pipelines, Risk Engine, Regulatory Compliance, Event-Driven Architecture, Distributed Systems, Fault-Tolerant Systems, Fault Tolerance, Idempotent Delivery, Scalable Architecture, Data Pipelines

Frontend & Mobile: React.js, Android SDK, Spring Boot, HTML/CSS

Developer Tools: Git, GitHub, GitHub Actions, CI/CD, Linux, Version Control, pytest, Unit Testing, Agile, SDLC, Object-Oriented Design, Cross-Functional Collaboration

Data & Analytics: NumPy, Pandas, PyTorch, Scikit-learn, SQL, NLP, RAG, Statistical Modeling, Statistical Analysis, Probability, Data Analysis, Financial Modeling, Risk Analysis, Actuarial Data

EXPERIENCE

Leidos

May 2025 - Present

Software Engineer Intern

- Designed scalable Java/Kafka pipeline with **at-least-once delivery semantics**, fault-tolerant streaming, and compliance-grade operational constraints for mission-critical satellite data
- Architected distributed microservices via dynamic programming: decoupled scheduling + pipeline + monitoring with **data integrity guarantees** and low-latency delivery for downstream satellite operations
- Deployed Python-tested Docker containers on Kubernetes via **GitHub Actions CI/CD** with comprehensive unit and integration test coverage

The Kendzierski Lab at the University of Wisconsin-Madison

September 2025 - Present

Data / Research Engineer Intern

- Engineered genomics pipeline optimization framework across **5+ scientific libraries**; reduced GPU processing costs by **300%** via systematic profiling, benchmarking, and throughput optimization
- Built TransferAgent automating NLP data synthesis and cross-document scoring; cut downstream pipeline processing time by **~60%**
- Designed event-driven data pipeline architecture for automated model selection across research stack; statistical analysis of throughput gains validated across all library configurations

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Trained and tuned ML classifiers (Decision Tree, KNN) achieving **93% classification accuracy** via hyperparameter tuning with cross-validated statistical significance; published **methodology notebook**
- Published "Early Skin Cancer Detection Improvement" through Inspirit AI peer-review; separately trained Random Forest on miRNA achieving **95% accuracy**

TECHNICAL PROJECTS

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Built object-oriented financial risk engine across **45M actuarial records** on **1,080-node ICD-10 graph** via SQL; outputs risk-adjusted, patient-level financial risk score from MEPS + CMS data
- NLP layer strictly separates inference from deterministic financial logic; outputs are **ACID-compliant, hallucination-free, and fully auditable** — production-grade financial system design

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Built failure-driven LLM agent self-improvement loop across **5,000+ tasks**, outperforming ReAct by **15pp**; provider-agnostic serving architecture with GitHub Actions CI/CD

LLM-Based SAT Practice Test Generator (<https://github.com/AkashGogate/SATPracticeTestGenerator>)

Jun 2024

- Built full-stack app (TypeScript/React.js + Node.js/FastAPI); GPT-4o adaptive test generation via REST API with agile session state, answer validation, configurable difficulty, and per-session score tracking

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics, client scheduling, and conflict resolution; coached athletes of all skill levels; managed all end-to-end client communications

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ loyal clients**; manage all operations, pricing, inventory, and client consultations